DATA STORYTELLING REPORT

Predicting Customer Churn with a Three-Model Stacked Ensemble

How a Recurrent Neural Network, a Discrete-Time Hazard Model, and a Survival Analysis layer were combined through a logistic-regression meta-learner to flag at-risk telecom subscribers before they leave.

100,000	**3**	**80.3%**	**33.1%**
CUSTOMER RECORDS	BASE MODELS STACKED	FINAL ROC-AUC	BASELINE CHURN RATE

Telecommunications · Customer Retention Analytics

Prepared from Telecom_Churn_Prediction.ipynb

WHY THIS MATTERS

One in Three Customers Is Walking Out the Door

Customer churn is quietly one of the most expensive problems in telecom: acquiring a new subscriber costs far more than keeping an existing one. This project set out to answer a concrete question — **can we reliably identify which customers are about to churn, and when?** The dataset covers 100,000 customers with demographic, contract, billing, and tenure information.

100,000
TOTAL CUSTOMERS

33,144
CHURNED (33.1%)

66,856
RETAINED (66.9%)

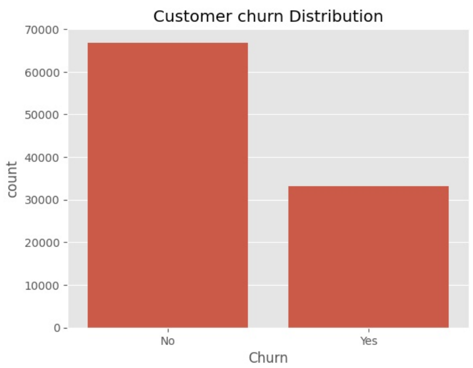

Fig 1. Customer churn distribution — a moderately imbalanced target

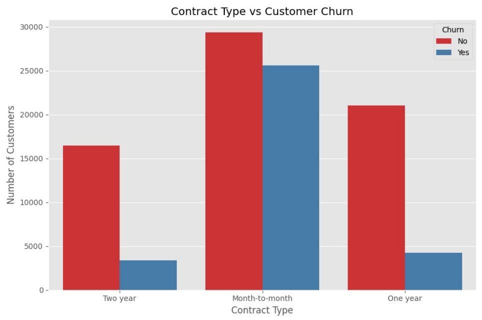

Fig 2. Month-to-month contracts churn far more than annual plans

Key signal: Contract type stands out immediately as one of the strongest churn drivers — customers on flexible month-to-month plans churn at a dramatically higher rate than those locked into one- or two-year contracts, hinting that contract commitment itself reduces churn risk.

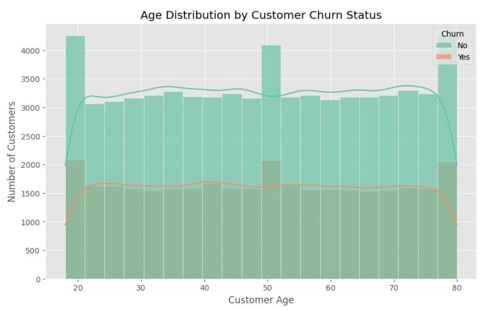

Fig 3. Age distribution split by churn status

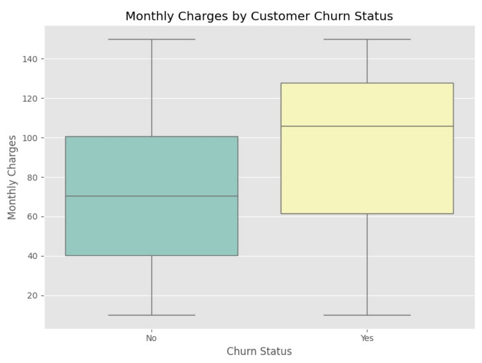

Fig 4. Monthly charges by churn status

Before modeling, the data was checked for missing values and duplicates — the dataset was clean on both counts. Age, tenure, monthly charges, and total charges were standardized, while gender, contract type, and payment method were one-hot encoded, and the data was split 70/10/20 into training, validation, and test sets.

THE APPROACH

Three Different Lenses on the Same Problem

Rather than relying on a single algorithm, this project builds **three conceptually different models** — each capturing a different aspect of churn behavior — and then lets a **meta-learner** decide how to weigh their opinions together. This "stacking" strategy typically outperforms any single model alone.

① RNN
Sequential deep-learning classifier over customer features

\+

② Hazard Model
Discrete-time probability of churning within a tenure interval

\+

③ Survival Model
Cumulative probability a customer "survives" over time

↓

④ Meta-Learner
Logistic Regression stacks all three signals into one final prediction

Pattern Recognition
RNN LEARNS NON-LINEAR FEATURE INTERACTIONS

Time-Based Risk
HAZARD MODEL ISOLATES RISK BY TENURE STAGE

Longevity View
SURVIVAL MODEL ESTIMATES "TIME TO CHURN"

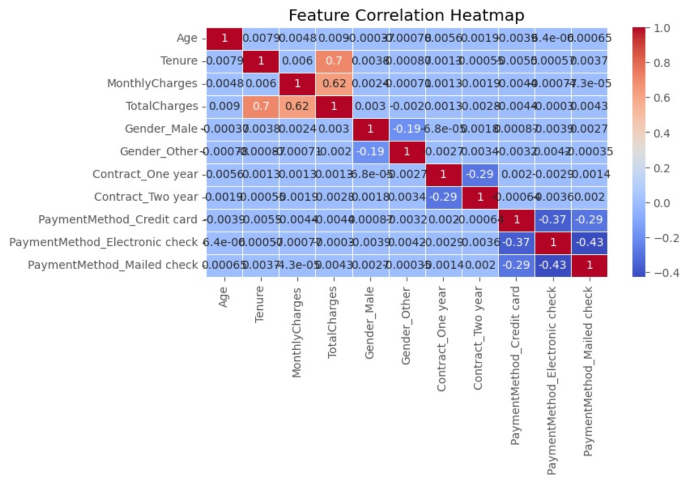

Fig 5. Feature correlation heatmap used to sanity-check inputs before modeling

The next three sections walk through each base model individually, followed by how their outputs were fused into the final stacked ensemble.

MODEL 1 — DEEP LEARNING

Teaching a Neural Network to Spot Risk Patterns

Each customer's feature set was reshaped into a sequence (one time step per feature) and fed into a **Simple RNN** — 64 units, a 30% dropout layer for regularization, and a 32-unit dense layer before a sigmoid output. The model was trained with early stopping and checkpointing to prevent overfitting, running up to 50 epochs.

SimpleRNN (64 units, tanh) · **Dropout 0.30** · **Dense (32, relu)** · **Sigmoid output** · **Adam optimizer** · **Early stopping · patience 10**

76.03% TEST ACCURACY

0.803 TEST ROC-AUC

0.461 TEST LOSS

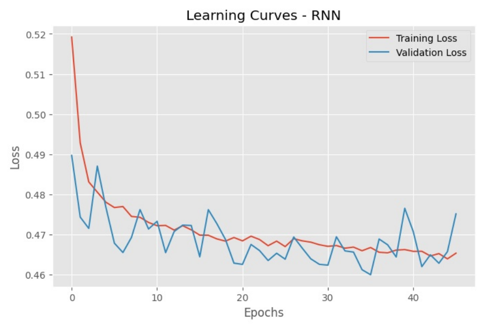

Fig 6. Training vs. validation loss — converges without overfitting

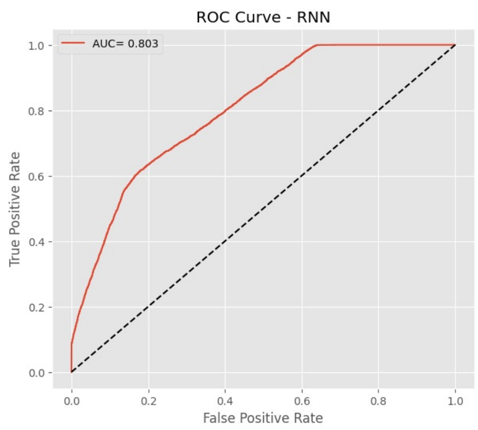

Fig 7. ROC curve for the standalone RNN (AUC 0.803)

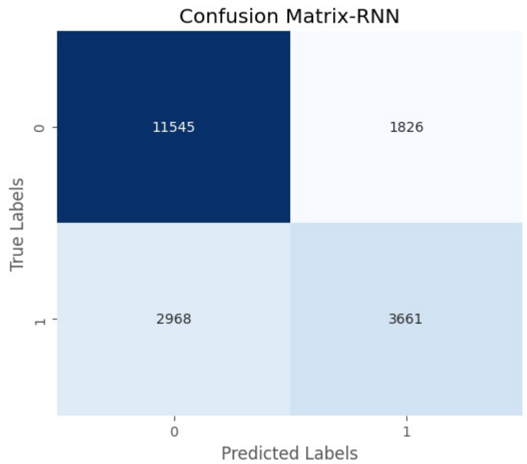

Fig 8. Confusion matrix — the RNN catches the majority of non-churners, but is more conservative on churners

Reading the results: The RNN reaches 67% precision and 55% recall on the churn class — solid pattern recognition on its own, but it is more cautious about flagging churners, which is exactly the gap the hazard and survival signals are brought in to close.

MODEL 2 — TIME-BASED RISK

When in a Customer's Lifecycle Is Risk Highest?

Tenure was split into four intervals (Year 1 through Year 4), and for each interval the **hazard probability** — the share of customers who churned within that specific window — was computed directly from the data. This produces a simple, interpretable time-based risk score that complements the RNN's feature-based view.

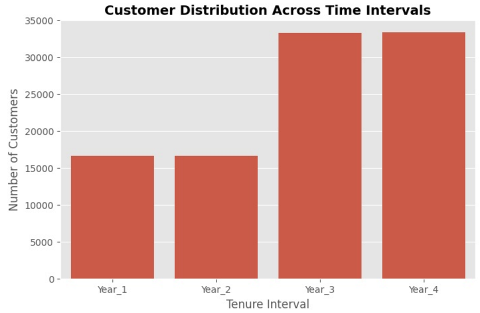

Fig 9. Customers are fairly evenly spread across tenure intervals

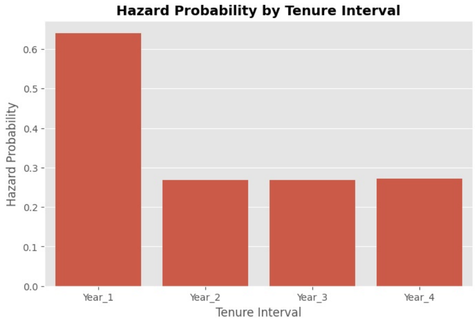

Fig 10. Hazard probability spikes sharply in Year 1

Tenure Interval	Customers	Churned	Hazard Probability
Year 1 (0–12 mo)	16,647	10,655	**64.0%**
Year 2 (12–24 mo)	16,637	4,471	26.9%
Year 3 (24–48 mo)	33,332	8,952	26.9%
Year 4 (48–72 mo)	33,384	9,066	27.2%

The headline finding: A new customer's risk of churning in their **first year (64.0%)** is roughly **2.4× higher** than in any later stage of the relationship. Once a customer survives past Year 1, their churn risk drops sharply and then stays fairly stable — the first 12 months are clearly the make-or-break window for retention efforts.

MODEL 3 — LONGEVITY VIEW

How Long Do Customers Really Stick Around?

Building on the hazard estimates, a cumulative **survival probability** was calculated for each tenure interval — the likelihood that a customer remains subscribed all the way through that point in time. This reframes churn as a "customer lifespan" problem rather than a single yes/no classification.

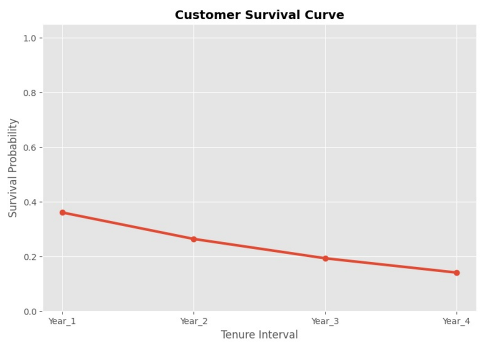

Fig 11. Customer survival curve across tenure intervals

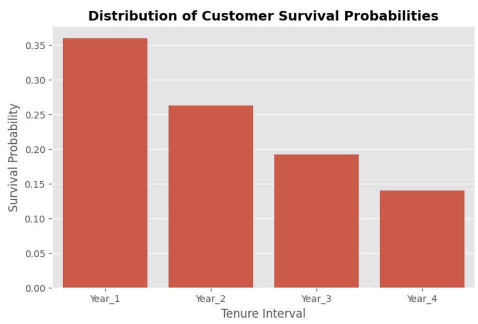

Fig 12. Survival probability drops fastest early, then flattens

36.0%
SURVIVE PAST YEAR 1

19.3%
SURVIVE PAST YEAR 3

14.0%
SURVIVE PAST YEAR 4

> **"Estimated median time-to-churn falls within the very first tenure interval — fewer than half of all customers make it past Year 1 without churning."**

The hazard and survival probabilities were then mapped back onto every customer record, turning a population-level analysis into two individual-level features ready to feed the stacking model — alongside the RNN's own churn probability for each customer.

META-LEARNER

Combining Three Opinions Into One Verdict

Each customer now carries three independent risk signals: the **RNN probability**, the **hazard probability**, and the **survival probability**. These three columns become the entire feature set for a final **Logistic Regression meta-learner** (class-balanced), trained on the validation set and evaluated on the held-out test set — a classic stacking architecture.

3
META-FEATURES IN

10,000
META-LEARNER TRAINING ROWS

20,000
HELD-OUT TEST ROWS

Why this works: The RNN captures complex feature interactions, the hazard model captures "how risky is this exact life-stage," and the survival model captures "how much runway does this customer have left." Logistic regression then simply learns the optimal weighted blend of these three complementary views.

THRESHOLD OPTIMIZATION

Rather than defaulting to a 0.50 cutoff, thresholds from 0.10 to 0.90 were swept to find the point that maximizes F1 score — balancing false alarms against missed churners.

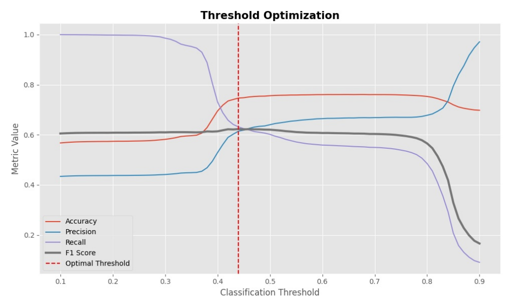

Fig 13. Accuracy, precision, recall and F1 across candidate thresholds

0.44
OPTIMAL DECISION THRESHOLD

0.623
F1 SCORE AT OPTIMAL THRESHOLD

THE PAYOFF

Does the Stacked Ensemble Deliver?

At the optimized 0.44 threshold, the stacking ensemble was evaluated on the 20,000-customer test set — data none of the three base models or the meta-learner had seen in this configuration before.

74.6% ACCURACY	**80.3%** ROC-AUC	**62.3%** F1 SCORE (CHURN CLASS)

Metric	Value	What It Means
Precision (Churn)	61.4%	Of customers flagged as "at risk," 61% truly churn
Recall (Churn)	63.1%	The model catches 63% of all customers who actually churn
F1 Score (Churn)	62.3%	Balanced precision/recall trade-off
ROC-AUC	80.3%	Strong ability to rank at-risk customers above safe ones

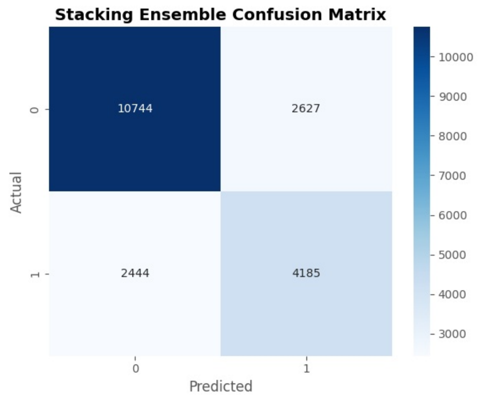

Fig 14. Confusion matrix — stacking ensemble, threshold 0.44

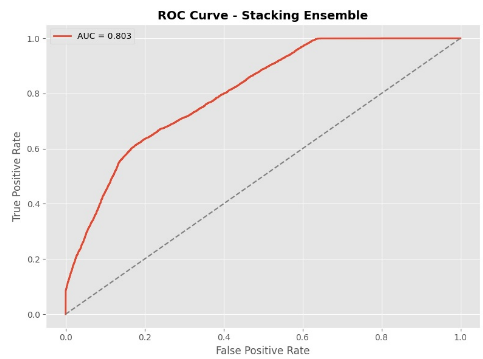

Fig 15. ROC curve — stacking ensemble (AUC 0.803)

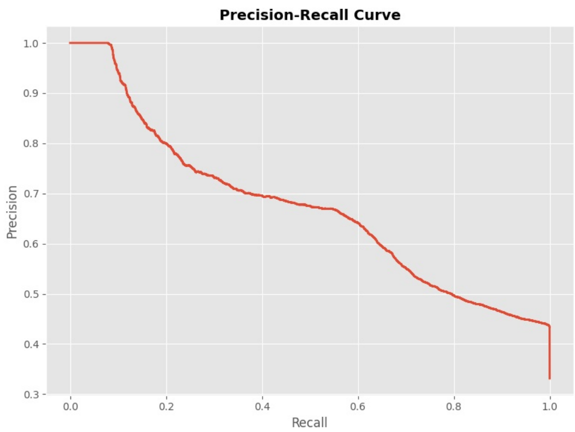

Fig 16. Precision-recall curve for the final stacked model

BRINGING IT TOGETHER

What the Business Should Do With This

1 Year 1 is the retention battleground.

With a 64% hazard rate in the first 12 months, onboarding and early-tenure engagement programs will have the highest return on retention spend.

2 Month-to-month plans are a red flag.

Flexible contracts correlate strongly with churn — incentivizing a move to annual contracts (loyalty pricing, bundled perks) is a lever worth testing.

3 Stacking beat any single model.

Blending pattern-based (RNN), time-based (hazard), and longevity-based (survival) signals lifted ranking quality to 0.803 ROC-AUC — a more complete picture than any one lens alone.

4 Recall still has room to grow.

At 63% recall, roughly a third of true churners still slip through — a good next step is testing tree-based meta-learners (e.g. gradient boosting) or adding more granular engagement features.

5 Threshold choice is a business decision.

Lowering the 0.44 cutoff would catch more churners at the cost of more false alarms — this dial should be tuned to match the actual cost of a retention offer versus a lost customer.

6 Actionable scoring, not just accuracy.

Because the model outputs a probability, customers can be ranked and prioritized for outreach rather than treated as a single pass/fail bucket.

Bottom line: a three-model stacked ensemble lifts churn detection to 80.3% ROC-AUC and gives the business both a "who" (via the RNN) and a "when" (via hazard & survival) — turning churn prediction into a targetable retention strategy rather than a guessing game.